MILITARY SPECIFICATION

BATTERY, PRIMARY, RESERVE, AMMONIA,
5 MILLIAMPERE HOUR CAPACITY

This amendment forms a part of Military Specification MIL-B-48343(AR), dated 17 May 1978, and is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

* 3.3.2 Delete paragraph in its entirety and substitute:

"3.3.2 Environmental. The ampule shall be capable of withstanding exposure to the environments listed herein.

3.3.2.1 Temperature cycling. The ampule shall be capable of withstanding temperature cycling from minus sixty-five degrees Fahrenheit plus or minus five degrees Fahrenheit to plus one hundred and seventy degrees Fahrenheit plus five degrees Fahrenheit ($-65^{\circ}F \pm 5^{\circ}F$ to $+ 170^{\circ}F + 5^{\circ}F - 0^{\circ}F$) when tested as specified in 4.5.1.2.

3.3.2.2 Time and temperature to fragmentation. The ampule, after an initial stabilization at $160^{\circ}F$ for at least one hour, shall be exposed to an increasing temperature of $5^{\circ}F$ (at a rate not greater than $5^{\circ}F$ per 15 minute period). The ampule shall not be fragmentized at any temperature of $170^{\circ}F$ or below."

* 3.3.3 Delete paragraph in its entirety and substitute:

"3.3.3 Ammonia composition.

a. The ammonia used to fill the ampule shall be anhydrous and of 99.98% minimum purity.

b. The filled and sealed ampule shall not contain more than .10% water."

FSC 1345

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PAGE 4

3.8, Delete "twenty-five degrees centigrade plus five degrees centigrade" and substitute "twenty-five degrees centigrade plus or minus five degrees centigrade".

PAGE 6

* Under "Quantity" for "Battery, Primary Reserve" delete "336" and substitute "336/672".

* 4.3.2 Delete paragraph in its entirety and substitute the following:

"4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein and the following paragraph:

4.3.2.1 Battery assembly test. Battery assemblies listed in Table I shall be tested as described below (See Figure 3):

a. Thirty-six (36) batteries shall be subjected to the five (5) environmental tests in the sequence shown. After the environmental testing, the batteries shall be subjected to the electrical test which includes activation (para. 4.5.9.2), rise time (para. 4.5.9.3) and long time capacity (4.5.9.5). However, the batteries will not require rotation during activation. Twelve (12) each batteries shall be tested at the temperature listed in Table II of 4.4.3.7.4. If the test sample contains one (1) reject for activation, rise time or Section I of the long time discharge requirements, a second sample of 36 batteries shall be tested. The second sample shall have passed the above criteria providing there are zero (0) rejects in the sample. The test shall be rejected if the first sample or the combined sample contain two (2) on more rejects for activation, rise time or discharge requirements of 4.4.3.7.2.

b. Two hundred and twenty-five (225) shall be subjected to activation, rise time and long time capacity. Seventy-five (75) each batteries shall be tested at the three temperatures listed in Table II of 4.4.3.7.4. One (1) reject for the first sample or three (3) rejects for the combined sample shall be allowed for activation and rise time requirement or Section 1 discharge requirements of 4.4.3.7.2. Three (3) rejects for the first sample or four (4) rejects for the combined sample shall be allowed through Section II discharge requirement of 4.4.3.7.3.

c. Twenty-five (25) each batteries shall be tested for short time capacity at the three temperatures listed in the Table II of 4.4.3.7.4. One (1) reject shall be allowed for the requirement of 4.4.3.7 and one (1) when tested as specified in 4.5.9.4."

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* 4.3.3 Delete paragraph in its entirety and substitute the following:

"4.3.3 Rejection. See MIL-A-48078. The First Article sample shall be rejected if any part, sub-assembly or assembly fails to comply with any of the examinations or test specified in this specification or any of the criteria specified in 4.3.2.1 above."

PAGE 6A

* Add new specification Page 6A as attached.

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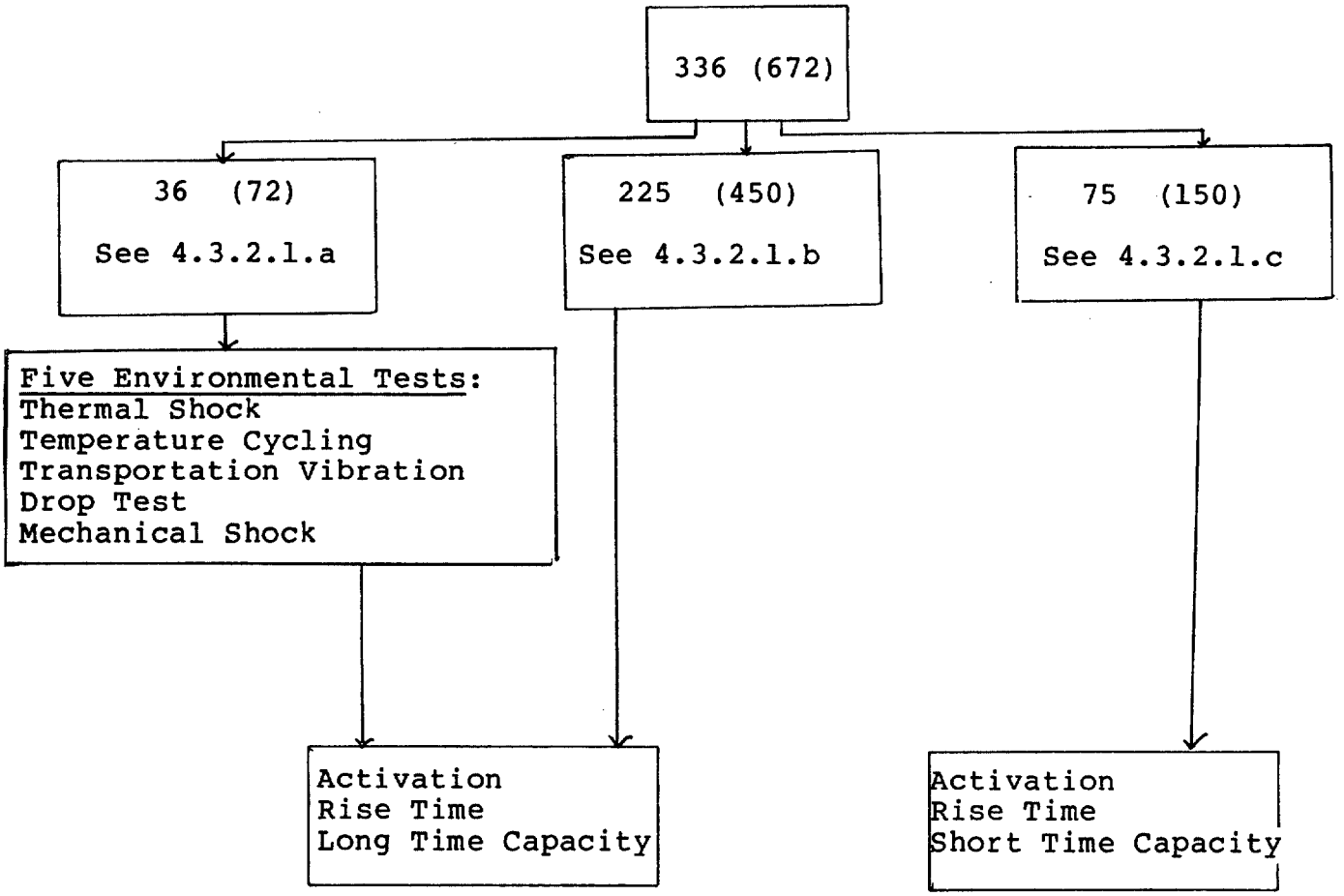


FIGURE 3. First Article Inspection Test Sequence for Battery Assembly.

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-48343(AR)
AMENDMENT 2

PARAGRAPH	TITLE		SHEET 3 OF 4		DRAWING NUMBER
	Battery, Primary Reserve Ammonia 5 Milliampere Hour Capacity				9298625
					NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cathode (Dwg. 9298640)				
	Moisture Content	25		3.4.1	4.5.2.1
	Weight	25		3.4.2	4.5.2.2
	Composition	25		3.2	4.5.2.3
	Cup and Anode Assembly (Dwg. 9298642)				
	Weld Test	25		3.2	4.5.5.1
	Vacuum Pull Test	25		3.2	4.5.5.2
	Cell Cup (Dwg. 9298625)				
	Examination for Defects	25		3.2	4.4.2.12
	Case Seal and Pre-initiation Voltage	25		3.5.1	4.5.9.1

NOTES:

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 4 OF 4		DRAWING NUMBER
	Battery, Primary Reserve Ammonia 5 Milliampere Hour Capacity				9298625
					NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Support Ampul (Dwg. 9298650)				
	Examination for Defects	25		3.2	4.4.2.13
	Hardness Test	Note a			
	Battery Assembly (Dwg. 9298625)	336 Note	b		
	Thermal Shock	36		3.6.1	4.5.10.1/4.3.2.a
	Temperature Cycling	36		3.6.2	4.5.10.2/4.3.2.a
	Transportation Vibration	36		3.6.3	4.5.10.3/4.3.2.a
	Drop Test	36		3.6.4	4.5.10.4/4.3.2.a
	Mechanical Shock	36		3.6.5	4.5.10.5/4.3.2.a
	Activation	225		3.5.2	4.5.9.2/4.3.2.b
	Rise Time	225		3.5.3	4.5.9.3/4.3.2.b
	Long-Time Capacity	225		3.5.5	4.5.9.5/4.3.2.b
	Short-Time capacity	75		3.5.4	4.5.9.4/4.3.2.c

NOTES: a. A coupon from the support ampule batch shall be used as the sample.
b. Double sample may apply. See 4.3.2.

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Table I: Delete "Sheet 1 of 4" and substitute "Sheet 1 of 5".

* Table I: Delete "Sheet 1 of 5" and substitute "Sheet 1 of 4".*

Under ampule glass, delete title, examinations and test criteria in its entirety and substitute:

"Examination or Test	No. of Sample Units	AQL or 100%	Req. Para	Para Ref/ Insp Method
Ampule Assy, Glass (Dwg. 9298628)				
Examination for defects	25	-	3.2	4.4.2.3
Seal	25	-	3.3.1	4.5.1.1
Ammonia Composition	25	-	3.2	4.5.1.3
Weight	25	-	3.2	4.5.1.4
Temperature Cycling (Short Term)	25	-	3.3.2.1	4.5.1.2.3
Temperature Cycling (Long Term)	100	-	3.3.2.1	4.5.1.2.1
Time & Temperature to Fragmentation	25	-	3.3.2.2	4.5.1.2.2"

PAGE 8

Table I: Delete "Sheet 2 of 4" and substitute "Sheet 2 of 5".

* Table I: Delete "Sheet 2 of 5" and substitute "Sheet 2 of 4".

PAGE 9

Table I: Delete "Sheet 3 of 4" and substitute "Sheet 3 of 5" as attached.

* Delete Page 9 in its entirety and substitute new page 9 as attached.

PAGE 10

Delete in its entirety and substitute new Page 10, as attached.

* Delete page 10 in its entirety and substitute new page 10 as attached.

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PAGE 10A

* Delete page 10A in its entirety.

PAGE 14

* 4.4.2.3 Add the following:

"Major	Exam/ Test	No. of Sample	Req Para	Para Ref/ Insp Method
108	Ammonia Weight	4.4.3.8	3.2	4.5.1.4"

PAGE 23

4.4.2.12, Major 106, change "4.5.9.5" to "4.5.9.5/4.4.3.7.2". Major 107, change "4.4.3.7.2" to "4.4.3.7.3".

PAGE 28

4.4.3.7.3, Change paragraph 4 from "Test temperature and the numbering samples per temperature group shall be in accordance with 4.4.3.7.4" to "Test temperature and the number of samples per temperature group shall be in accordance with the requirements of 4.4.3.7.4."

PAGE 29

4.4.3.7.4, Table III: Change "For sample size of 200 batteries" to "For sample size of 175 batteries".

PAGE 30

4.4.4, Delete in its entirety and substitute "4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit inspection equipment designs for approval in accordance with the terms of the contract. See Section 6 of MIL-A-48078."

PAGE 31

* Delete paragraph 4.5.1.2 in its entirety and substitute:

"4.5.1.2 Environmental.

a. The capability of the ampule(s) to withstand exposure to the environments shall be substantiated by an evaluation of their performance or condition prior to, during, and/or after exposure to the environments specified in the table of inspections.

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b. In addition, to the tests or examinations after the environmental exposure(s) specified in the tables of inspection, the ampules shall be visually examined to assure that they have survived the exposure(s) without physical damage that would impair their form or fit.

4.5.1.2.1 Temperature cycling (long term). The ampules shall be tested as specified in MIL-STD-331, Test 105. The following exceptions and details shall apply:

a. Humidity requirement of MIL-STD-331 not required.

b. Method of Mounting: The ampule(s) may be mounted in any suitable fixture that will provide isolation from one to another and allow appropriate heat transfer to the ampule(s). The isolation shall be sufficient to prevent adjacent ampule(s) from self destructing or damage from shock and/or fragmentation from other failing ampule(s) in the same test.

c. Measurements after test: See 4.5.1.2b.

4.5.1.2.2 Time and temperature to fragmentation.

a. Mount the components as specified in 4.5.1.2.

b. Control the test temperature as indicated in 3.3.2.2.

c. Determine time and temperature of fragmentation.

d. Calculate results for conformance to 3.3.2.2.

e. Use caution when testing.

4.5.1.2.3 Temperature cycling (short term). The ampule shall be temperature cycled as specified when mounted in an isolated fixture in the temperature chambler. The ampule shall be cycled five (5) times as follows:

a. Heat soaked for a minimum of one and five tenths (1.5) hours at a minimum temperature of plus one hundred and seventy degrees fahrenheit (+170°F).

b. The temperature shall be gradually reduced.

c. The ampule shall be cold soaked for a minimum of one (1) hour at a maximum temperature of minus sixty-five degrees fahrenheit (-65°F).

d. This is the completion of one (1) cycle.

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e. The ampule temperature shall be gradually raised and the start of the second cycle and each successive cycle shall commense. Each cycle shall be in accordance with steps a through c. The five cycles shall be completed within seventy-two (72) hours.

Any ampule that cracks, ruptures, discolors or leaks shall be classed defective and removed from the lot. (Non-destructive test)."

PAGE 34

4.5.9.2, Last subparagraph: Change "5 1/4C" to "5°C".

PAGE 35

4.5.9.5, Change "During first article acceptance testing the contractor shall continuously monitor the battery voltage until it drops below 2.7 volts" to "During first article acceptance testing, the contractor shall monitor the battery voltage a minimum of once an hour until it drops below two and seven tenths (2.7) volts".

4.5.10.3, Change "Test Method 104 of MIL-STD-331" to "Test Method 119 of MIL-STD-331 shall be used except test vehicle shall be randomly located in standard shipping pack."

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes to liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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